

Score: / 10

Calculus I (MATH 2200)

Spring 2020

Wednesday, 05/06/2020

Quiz 13: Section 5.3, 5.4

Name (Print): _____

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Section 5.3

2 pts

1. Evaluate the definite integral $\int_0^{10} (60x - 6x^2) \, dx$ using the Fundamental Theorem of Calculus.

(A) 5000

(B) 1000

(C) 0

(D) 100

Using the antiderivative rules of Section 4.9, an antiderivative of $60x - 6x^2$ is $30x^2 - 2x^3$. By the Fundamental Theorem, the value of the definite integral is

$$\begin{aligned}\int_0^{10} (60x - 6x^2) \, dx &= (30x^2 - 2x^3) \Big|_0^{10} \\ &= (30(10)^2 - 2(10)^3) - (30(0)^2 - 2(0)^3) \\ &= (3000 - 2000) - 0 \\ &= 1000.\end{aligned}$$

2 pts

2. Evaluate the definite integral $\int_0^{2\pi} 3\sin(x) \, dx$ using the Fundamental Theorem of Calculus.

(A) 3

(B) 6π

(C) 0

(D) -3

Using the antiderivative rules of Section 4.9, an antiderivative of $3\sin(x)$ is $-3\cos(x)$. By the Fundamental Theorem, the value of the definite integral is

$$\begin{aligned}\int_0^{2\pi} 3\sin(x) \, dx &= -3\cos(x) \Big|_0^{2\pi} \\ &= (-3\cos(2\pi)) - (-3\cos(0)) \\ &= -3 - (-3) \\ &= 0.\end{aligned}$$

2 pts

3. Use the Fundamental Theorem of Calculus to simplify the expression

$$\frac{d}{dx} \int_1^x \sin^2(t) \, dt.$$

Ⓐ $1 + \cos^2(x)$

Ⓑ $\cos^2(x)$

Ⓒ $-\sin(x) \cos(x)$

Ⓓ $\sin^2(x)$

Using Part *I* of the Fundamental Theorem of Calculus, we see that

$$\frac{d}{dx} \int_1^x \sin^2(t) \, dt = \sin^2(x).$$

Section 5.4

3 pts

1. A hiking trail has an elevation given by

$$f(x) = 60x^3 - 650x^2 + 1200x + 4500,$$

where f is measured in feet above sea level and x represents horizontal distance along the trail in miles, with $0 \leq x \leq 5$. Approximately, what is the average elevation of the trail?

- (A) 3958.33 ft (B) 11021.67 ft (C) 291.33 ft (D) 11875.00 ft

If one graphs the function, we may notice that the trail ranges between elevations of about 2000 and 5000 ft. If we let the endpoints of the trail correspond to the horizontal distances $a = 0$ and $b = 5$, the average elevation of the trail in feet is

$$\begin{aligned}\bar{f} &= \frac{1}{b-a} \int_a^b f(x) \, dx \\ &= \frac{1}{5-0} \int_0^5 (60x^3 - 650x^2 + 1200x + 4500) \, dx \\ &= \frac{1}{5} \left(\frac{60}{4}x^4 - \frac{650}{3}x^3 + \frac{1200}{2}x^2 + 4500x \right) \Big|_0^5 \\ &= \frac{11875}{3} \approx 3958.33\end{aligned}$$

So, the average elevation of the trail is slightly less than 3960 ft.

1 pts

2. Find the point(s) on the interval $[0, 1]$ at which $f(x) = 2x$ equals its average value on $[0, 1]$.

Ⓐ $x = \frac{1}{2}$

Ⓑ $x = 0$

Ⓒ $x = -2$

Ⓓ $x = 1$

The average value of f on $[0, 1]$ is

$$\bar{f} = \frac{1}{1-0} \int_0^1 2x \, dx = x^2 \Big|_0^1 = 1.$$

Now, we must find the points on $[0, 1]$ at which $f(x) = 1$. Notice, this is equivalent to solving $f(x) - 1 = 0$, i.e. solving the equation $2x - 1 = 0$. Solving, we produce the solution

$$x = \frac{1}{2}.$$

So, for this values of x , we may conclude that f equals its average value on $[0, 1]$.